

RheumLens: a donor-level audit benchmark for single-cell foundation-model disease representations in systemic lupus erythematosus

Working draft v0.6 · 23 June 2026 · Target journal to be selected

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RheumLens: a donor-level audit benchmark for single-cell foundation-model disease representations in systemic lupus erythematosus

Running title: Donor-level audit of single-cell foundation models in SLE

Manuscript version: working v0.6, assembled 2026-06-23

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Abstract

Single-cell foundation models are increasingly used as reusable feature extractors for disease studies. Their donor-level value remains difficult to quantify because many evaluations mix cells from the same donor across train/test splits, omit strong expression baselines, or report high discrimination without testing covariate sensitivity. We built RheumLens, a donor-level benchmark and audit workflow for systemic lupus erythematosus (SLE) single-cell RNA-seq cohorts. The workflow uses donor-disjoint folds, out-of-fold prediction, source-only transfer, paired resampling, representation provenance checks, cell-count sensitivity, covariate sensitivity, and kernel diagnostics.

Across GSE135779 matched-500, GSE174188 CD4, and GSE285773 CD4 analyses, disease signal was strong and reproducible. In GSE135779, raw pseudobulk reached AUC 0.972, focus_lite@scGPT reached 0.967, and Geneformer mean reached 0.926 after full provenance repair. In GSE174188 CD4, raw pseudobulk reached AUC 0.988, donor expression PCA 0.986, donor mean HVG 0.984, scGPT mean 0.978, and kme_multiscale@scGPT 0.978. Source-only transfer between GSE285773 and GSE174188

showed cross-study signal: GSE285773→GSE174188 AUCs were 0.730 for scGPT mean, 0.790 for donor expression PCA, and 0.806 for donor mean HVG; GSE174188→GSE285773 AUCs were 0.881, 0.931, and 0.900, respectively. Covariate sensitivity was substantial: measured covariates alone achieved AUC 0.846 in GSE174188 CD4, while residualized scGPT mean, donor expression PCA, and donor mean HVG reached AUCs of 0.868, 0.853, and 0.870.

RheumLens shows that single-cell foundation embeddings recover reproducible SLE-associated donor signal, while expression baselines remain highly competitive. The strongest manuscript finding is the audit itself: high disease AUCs require donor-level validation, expression baselines, provenance controls, and covariate sensitivity before biological or model-superiority claims are made.

Keywords: single-cell RNA-seq; foundation model; scGPT; Geneformer; systemic lupus erythematosus; donor-level classification; benchmark; covariate sensitivity

1. Introduction

Single-cell foundation models convert transcriptomes into compact representations intended to transfer across tissues, cell types, and tasks. Models such as scGPT and Geneformer are now practical tools for cell-level embedding extraction, and their use in disease datasets is expanding. The disease setting creates a specific statistical problem: thousands of cells are measured, but the independent unit is usually the donor. Evaluations that split cells instead of donors can report inflated performance. Evaluations without conventional expression baselines can also mistake recoverable expression signal for foundation-model-specific value.

Systemic lupus erythematosus provides a demanding test case. SLE has a strong, reproducible interferon-associated transcriptional program and multiple public single-cell cohorts with case-control donors. These properties make it possible to test whether frozen embeddings recover disease signal, whether that signal transfers across studies, and whether more complex donor-level methods add information beyond straightforward expression summaries.

The study is positioned against two related lines of work. First, scGPT and Geneformer introduced reusable transcriptomic representations that can be extracted from frozen cell-level models. Second, recent single-cell foundation-model evaluations and broader perturbation-model benchmarks have emphasized that strong linear, PCA, pseudobulk, and other simple baselines often remain competitive. RheumLens therefore treats foundation embeddings as auditable representations rather than assumed improvements over expression summaries.

RheumLens was developed as a donor-level benchmark and audit framework for this setting. The project evaluates scGPT, Geneformer, expression pseudobulk, donor expression PCA, HVG baselines, KME-style

distribution embeddings, FOCUS-style methods, attention/MIL-style models, and covariate-adjusted sensitivity analyses. The final analysis emphasizes out-of-fold donor predictions, fixed folds, source-only transfer, provenance checks, and explicit claim boundaries.

The main contributions are: (i) a donor-disjoint SLE benchmark spanning matched pediatric data, adult CD4 data and pediatric CD4 transfer; (ii) a model-comparison audit that includes strong expression baselines, provenance repair, paired resampling, permutation tests and KME implementation diagnostics; (iii) covariate sensitivity analyses that expose the extent to which measured donor/sample variables can explain high disease AUC; and (iv) a reproducibility package with code, figures, supplementary tables, manifests and a public data-release plan.

2. Results

2.1 RheumLens assembled a donor-level benchmark spanning matched cells, large adult CD4 data, and pediatric CD4 transfer

The final analysis package contains the core processed data, embeddings, fold files, out-of-fold predictions, bootstrap summaries, and audit reports needed for manuscript assembly. GSE135779 matched-500 provides a balanced, controlled setting with 44 donors and 500 cells per donor. GSE174188 CD4 provides the largest adult CD4 case-control analysis with 261 donors. GSE285773 provides an independent pediatric CD4 cohort with 26 donors. All formal analyses use donor-level labels and donor-disjoint evaluation.

Figure 1 summarizes the leading benchmark results. The matched-500 and GSE174188 CD4 analyses both show high donor-level disease discrimination across multiple method families.

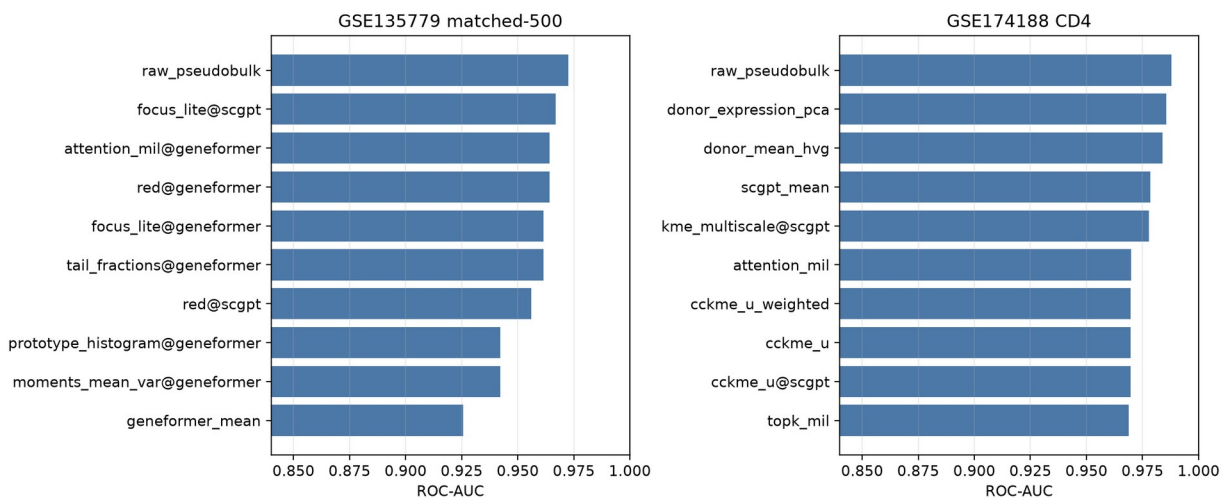


Figure 1. Top benchmark AUCs.

Table 1. Top benchmark results.

analysis	method_id	n_donors	roc_auc	pr_auc	brier
GSE135779 matched-500	raw_pseudobulk	44	0.972	0.991	0.192
GSE135779 matched-500	focus_lite@scgpt	44	0.967	0.99	0.088
GSE135779 matched-500	attention_mil@geneformer	44	0.964	0.99	0.089
GSE135779 matched-500	red@geneformer	44	0.964	0.989	0.106
GSE135779 matched-500	tail_fractions@geneformer	44	0.961	0.987	0.102
GSE135779 matched-500	focus_lite@geneformer	44	0.961	0.987	0.104
GSE135779 matched-500	red@scgpt	44	0.956	0.987	0.092
GSE135779 matched-500	prototype_histogram@geneformer	44	0.942	0.981	0.104
GSE135779 matched-500	moments_mean_var@geneformer	44	0.942	0.981	0.135
GSE135779 matched-500	moments_mean_var@scgpt	44	0.926	0.977	0.143
GSE135779 matched-500	geneformer_mean	44	0.926	0.974	0.157
GSE135779 matched-500	quantiles@geneformer	44	0.917	0.974	0.161
GSE174188 CD4	raw_pseudobulk	261	0.988	0.993	0.054
GSE174188 CD4	donor_expression_pca	261	0.986	0.992	0.054
GSE174188 CD4	donor_mean_hvg	261	0.984	0.991	0.055
GSE174188 CD4	scgpt_mean	261	0.978	0.988	0.062
GSE174188 CD4	kme_multiscale@scgpt	261	0.978	0.986	0.056
GSE174188 CD4	attention_mil	261	0.97	0.984	0.064
GSE174188 CD4	cckme_u_weighted	261	0.97	0.979	0.065
GSE174188 CD4	cckme_u@scgpt	261	0.97	0.978	0.064
GSE174188 CD4	cckme_u	261	0.97	0.978	0.064
GSE174188 CD4	topk_mil	261	0.969	0.982	0.066
GSE174188 CD4	moments_mean_var@scgpt	261	0.967	0.965	0.066
GSE174188 CD4	tail_fractions@scgp	261	0.962	0.968	0.064

analysis	method_id	n_donors	roc_auc	pr_auc	brier
	t				

2.2 Foundation embeddings recover SLE-associated donor signal, while expression baselines remain highly competitive

In GSE135779 matched-500, raw pseudobulk achieved AUC 0.972. scGPT-derived methods were also strong: focus_lite@scGPT reached 0.967 and scGPT mean reached 0.846. After Geneformer provenance repair, Geneformer mean reached AUC 0.926 and focus_lite@geneformer reached 0.961. This corrected the earlier failure mode caused by an unknown-provenance Geneformer NPZ and incompatible extraction environment.

The lower scGPT mean AUC in GSE135779 matched-500 was examined as a method-resolution finding rather than treated as a nuisance. Paired donor bootstrap showed that raw pseudobulk exceeded scGPT mean by ΔAUC 0.127 (95% CI 0.039 to 0.234; Holm-adjusted $P=0.032$), and focus_lite@scGPT exceeded scGPT mean by ΔAUC 0.121 (95% CI 0.036 to 0.226; Holm-adjusted $P=0.032$). RED@scGPT showed a similar numerical advantage that did not survive Holm correction. This pattern suggests that simple mean pooling can underuse donor-distribution structure in GSE135779, while the larger GSE174188 CD4 analysis is closer to an AUC ceiling where paired differences are small.

In GSE174188 CD4, raw pseudobulk achieved AUC 0.988, donor expression PCA 0.986, donor mean HVG 0.984, scGPT mean 0.978, and kme_multiscale@scGPT 0.978. These results show robust disease signal in frozen embeddings and direct expression summaries. Expression baselines were among the strongest methods in the final analysis.

Table 2. Primary method anchors.

analysis	method_id	n_donors	roc_auc	pr_auc	brier
GSE135779 matched-500	raw_pseudobulk	44	0.972	0.991	0.192
GSE135779 matched-500	focus_lite@scgpt	44	0.967	0.99	0.088
GSE135779 matched-500	attention_mil@geneformer	44	0.964	0.99	0.089
GSE135779 matched-500	geneformer_mean	44	0.926	0.974	0.157
GSE135779 matched-500	donor_expression_pca	44	0.915	0.972	0.112
GSE135779 matched-500	donor_mean_hvg	44	0.893	0.955	0.181

analysis	method_id	n_donors	roc_auc	pr_auc	brier
GSE135779 matched-500	kme_multiscale@scgpt	44	0.873	0.958	0.198
GSE135779 matched-500	scgpt_mean	44	0.846	0.951	0.164
GSE174188 CD4	raw_pseudobulk	261	0.988	0.993	0.054
GSE174188 CD4	donor_expression_pca	261	0.986	0.992	0.054
GSE174188 CD4	donor_mean_hvg	261	0.984	0.991	0.055
GSE174188 CD4	scgpt_mean	261	0.978	0.988	0.062
GSE174188 CD4	kme_multiscale@scgpt	261	0.978	0.986	0.056
GSE174188 CD4	attention_mil	261	0.97	0.984	0.064
GSE174188 CD4	focus_lite@scgpt	261	0.958	0.974	0.08

2.3 Source-only CD4 transfer supports cross-study signal

We repaired the local GSE174188 CD4 expression matrix metadata by recovering true gene symbols from the server-side h5ad file. After this repair, GSE174188 and GSE285773 shared 18,916 expression features, enabling expression-space transfer as well as scGPT transfer.

In source-only P9 transfer, training on GSE285773 and testing on GSE174188 CD4 produced AUC 0.730 for scGPT mean, 0.790 for donor expression PCA, and 0.806 for donor mean HVG. The reverse direction, training on GSE174188 CD4 and testing on GSE285773, produced AUC 0.881 for scGPT mean, 0.931 for donor expression PCA, and 0.900 for donor mean HVG. The reverse target has only 26 donors, so intervals are wide, but the signal is clear.

We then extended P9 to non-neural structured transfer methods. In GSE285773→GSE174188 CD4 transfer, expression baselines remained strongest: donor mean HVG reached AUC 0.806 and donor expression PCA 0.790, while the best structured scGPT method was focus_lite@scGPT at AUC 0.754. In the reverse GSE174188 CD4→GSE285773 transfer, moments_mean_var@scGPT reached AUC 0.963, focus_lite@scGPT reached AUC 0.931, and donor expression PCA also reached AUC 0.931. KME did not improve transfer in either direction. The 26-donor pediatric target makes this extension hypothesis-generating, but it addresses the method-coverage gap in the original P9 table.

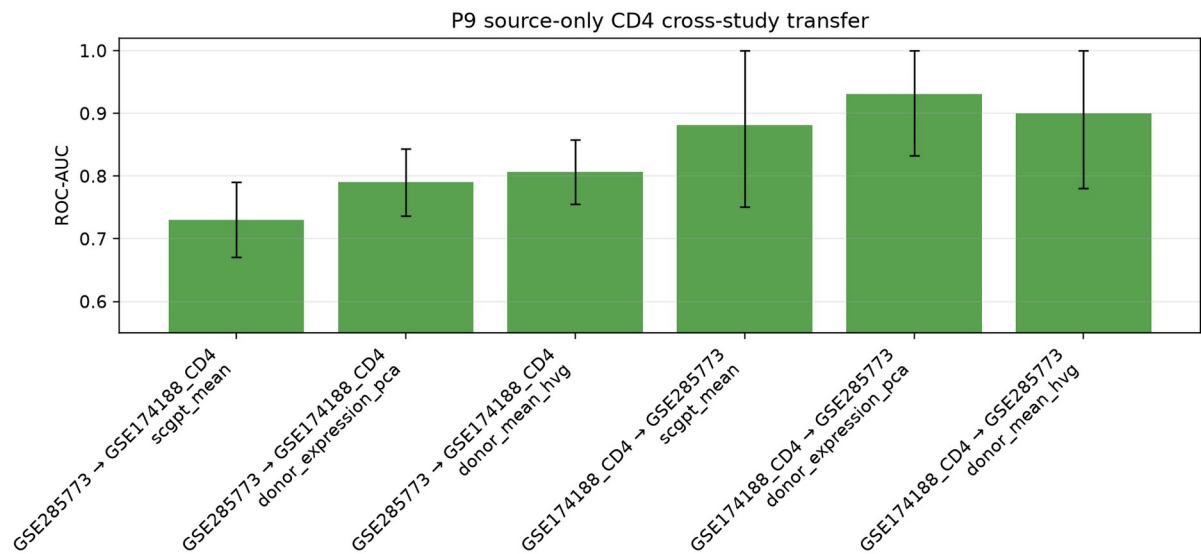


Figure 2. P9 source-only CD4 transfer.

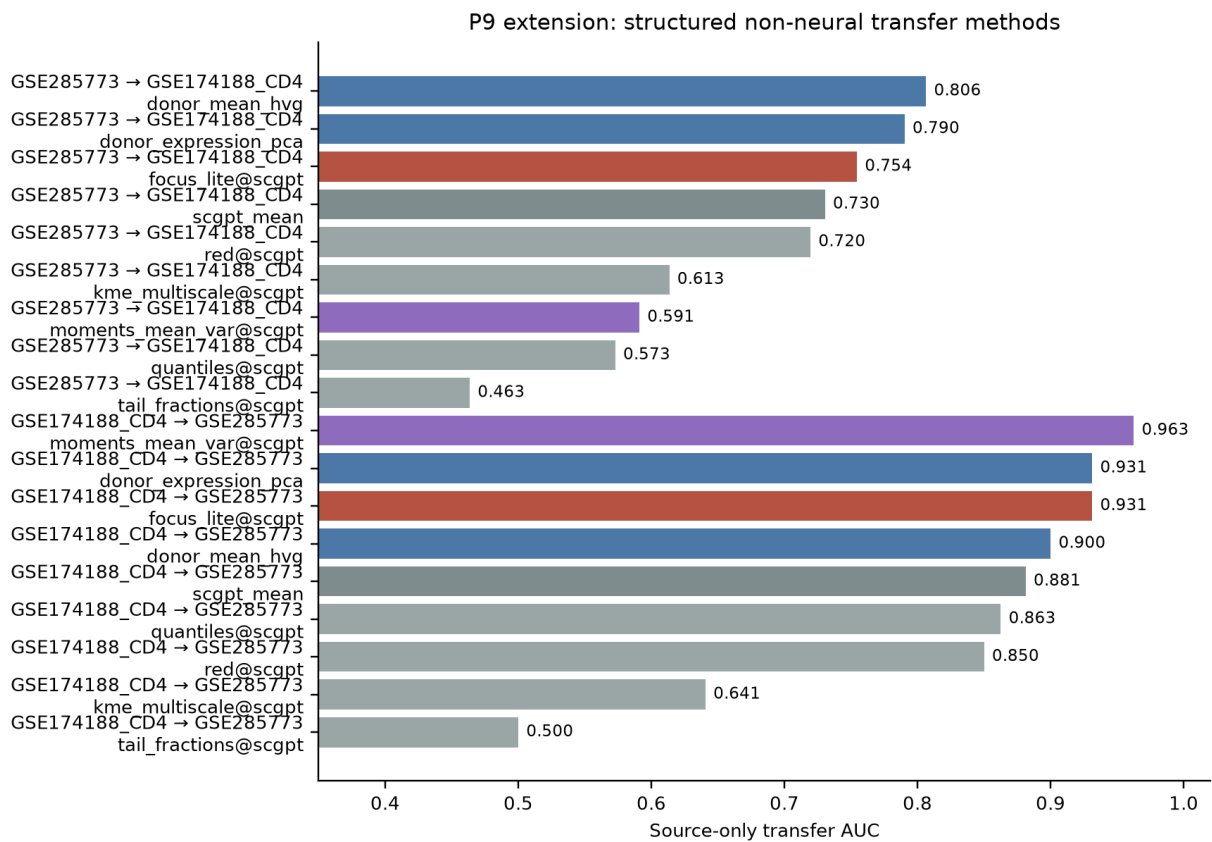


Figure 16. P9 structured source-only transfer extension.

Table 3. Source-only transfer results.

direction	method_id	n_donors	n_cases	n_controls	auc	auc_ci_low	auc_ci_high	pr_auc	pr_auc_ci_low	pr_auc_ci_high	brier
GSE285773_to_GSE174188_CD4	scgpt_mean	261	162	99	0.73	0.67	0.791	0.845	0.808	0.88	0.443
GSE285773_to_GSE174188_CD4	donor_expression_pca	261	162	99	0.79	0.737	0.844	0.896	0.868	0.924	0.252
GSE285773_to_GSE174188_CD4	donor_mean_hvg	261	162	99	0.806	0.755	0.858	0.893	0.862	0.922	0.313
GSE174188_CD4_to_GSE285773	scgpt_mean	26	16	10	0.881	0.75	1	0.935	0.856	0.99	0.615
GSE174188_CD4_to_GSE285773	donor_expression_pca	26	16	10	0.931	0.833	1	0.969	0.917	1	0.183
GSE174188_CD4_to_GSE285773	donor_mean_hvg	26	16	10	0.9	0.78	1	0.953	0.887	1	0.476

2.4 Repeated cross-validation and permutation testing confirmed that the primary signals are stable

Repeated cross-validation on GSE174188 CD4 produced tightly concentrated AUC distributions for the three primary methods. Across 30 repeats, mean AUC was 0.977 for scGPT mean, 0.986 for donor expression PCA, and 0.975 for kme_multiscale@scGPT. These values agreed with the fixed-fold P8.2 reference runs.

Formal donor-label permutation tests were available as full null distributions for donor expression PCA and kme_multiscale@scGPT. Both observed AUCs were far outside the 1,000-replicate null distributions, with empirical $P=0.000999$. This establishes that the primary GSE174188 CD4 signals are not artifacts of the fixed fold assignment.

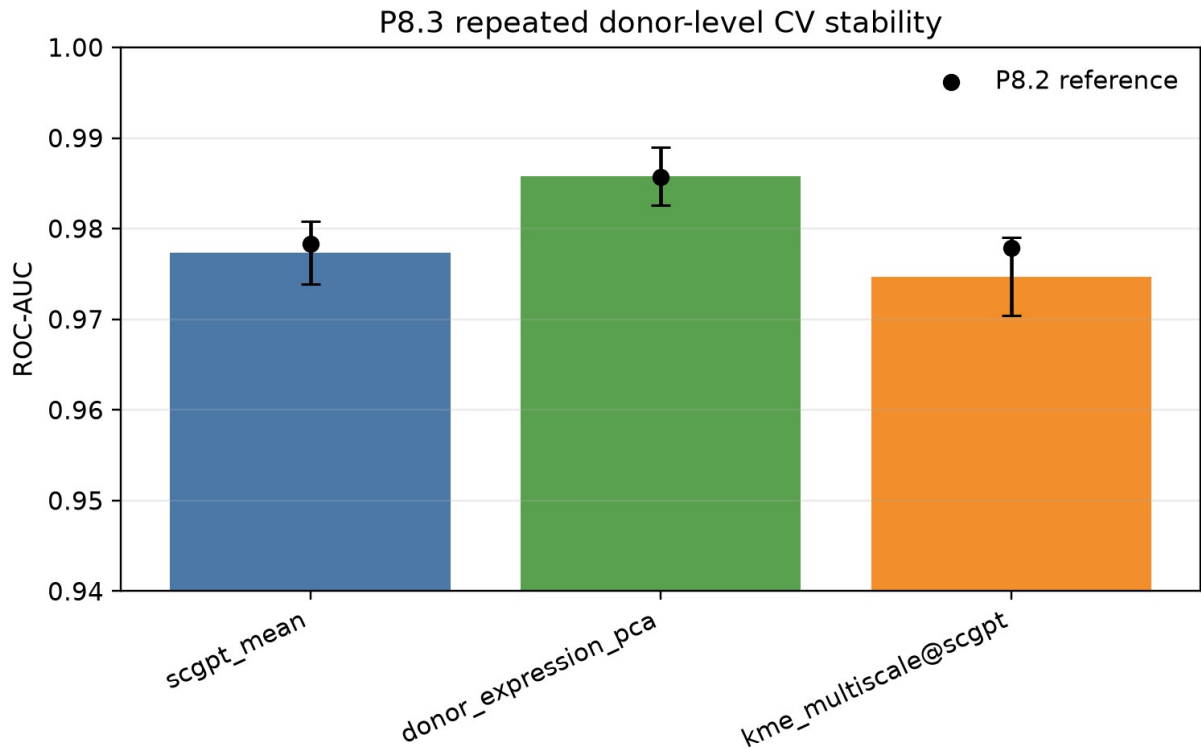


Figure 4. Repeated CV stability.

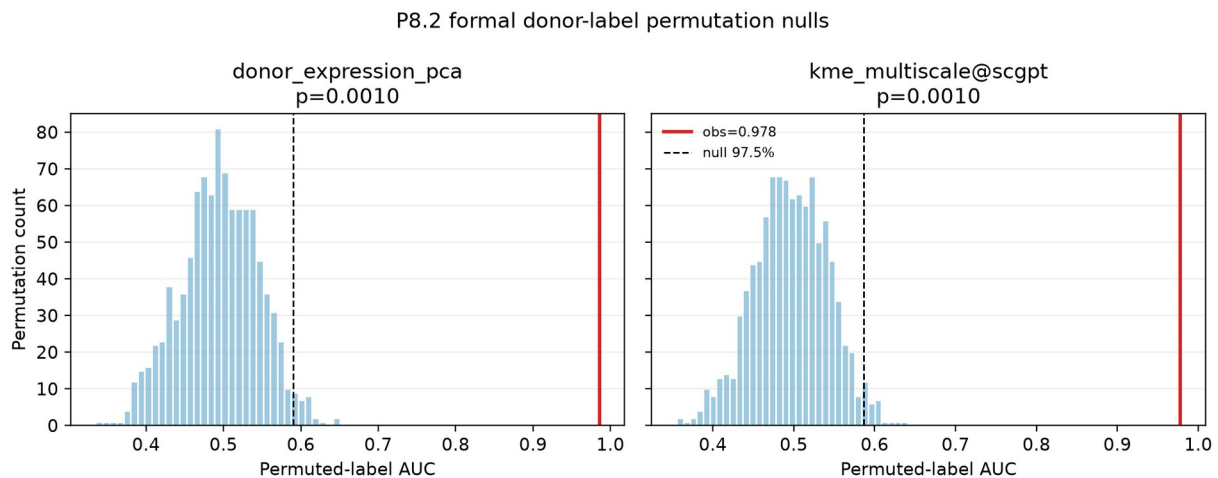


Figure 5. Formal permutation nulls.

2.5 Cell-count sensitivity confirms that the primary signal is stable across donor sampling levels

P8.4 tested common-support sampling across cell-count levels. The analysis used nested sampling manifests and a corrected V5 runner that invoked the registered method objects rather than collapsing all methods into mean pooling. At common-support level 50, the 10-repeat mean AUC was 0.953 for

scGPT mean, 0.970 for donor expression PCA, and 0.948 for kme_multiscale@scGPT. Across levels 25, 50, 100, 200, and 500, disease discrimination remained high. The result supports stability of the donor-level signal under cell-count perturbation.

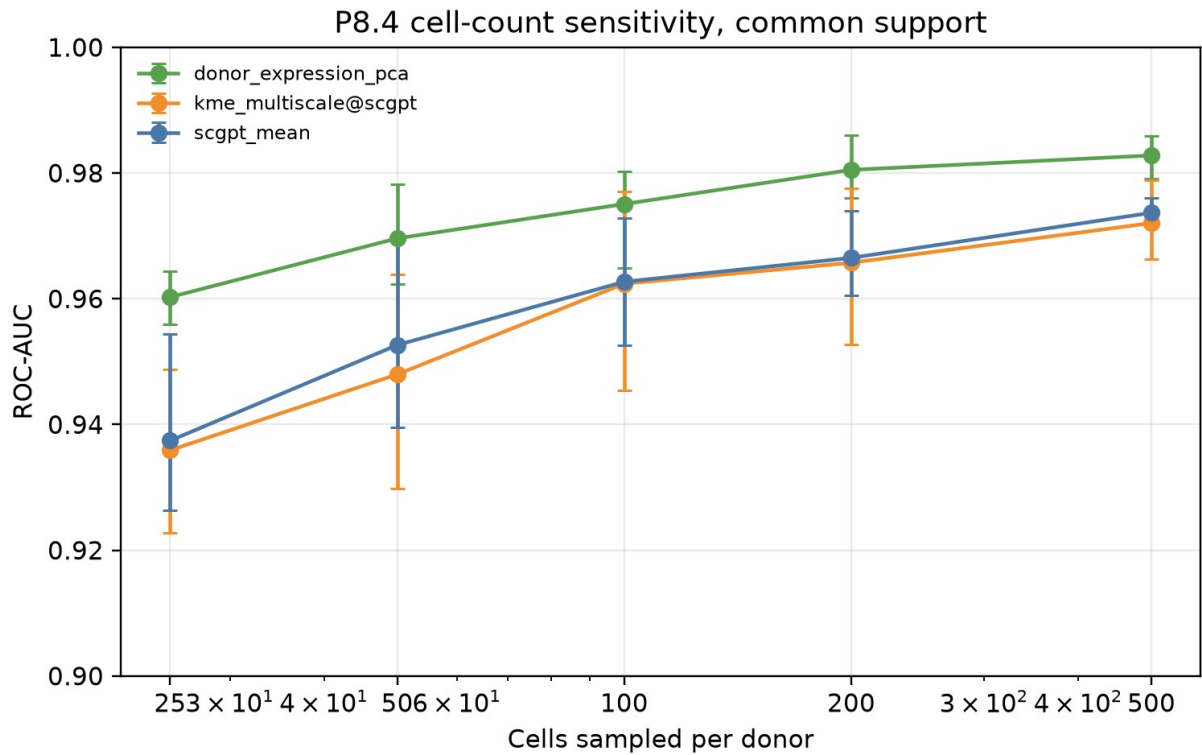


Figure 6. Cell-count sensitivity.

Table 4. Cell-count sensitivity summary.

estimand	level	method	n_repeats	mean_auc	sd_auc	min_auc	max_auc	mean_pr_auc
available_donor	25	donor_expression_pca	10	0.965	0.002	0.962	0.969	0.979
available_donor	25	kme_multiscale@scgpt	10	0.937	0.008	0.924	0.947	0.962
available_donor	25	scgpt_mean	10	0.944	0.008	0.933	0.958	0.967
available_donor	50	donor_expression_pca	10	0.974	0.003	0.969	0.978	0.984
available_donor	50	kme_multiscale@scgpt	10	0.95	0.01	0.93	0.965	0.969
available_donor	50	scgpt_mean	10	0.958	0.008	0.946	0.972	0.975
available_donor	100	donor_expression_pca	10	0.978	0.004	0.971	0.983	0.987

estimand	level	method	n_repeats	mean_auc	sd_auc	min_auc	max_auc	mean_pr_auc
available_donor	100	kme_multiscale@scgpt	10	0.961	0.01	0.944	0.977	0.976
available_donor	100	scgpt_mean	10	0.966	0.006	0.958	0.977	0.98
available_donor	200	donor_expression_pca	10	0.983	0.004	0.978	0.988	0.99
available_donor	200	kme_multiscale@scgpt	10	0.967	0.007	0.959	0.98	0.979
available_donor	200	scgpt_mean	10	0.97	0.004	0.964	0.977	0.982
available_donor	500	donor_expression_pca	10	0.983	0.003	0.979	0.986	0.989
available_donor	500	kme_multiscale@scgpt	10	0.972	0.004	0.965	0.979	0.982
available_donor	500	scgpt_mean	10	0.974	0.002	0.971	0.976	0.984
common_support	25	donor_expression_pca	10	0.96	0.003	0.956	0.965	0.974
common_support	25	kme_multiscale@scgpt	10	0.936	0.01	0.922	0.949	0.957
common_support	25	scgpt_mean	10	0.937	0.01	0.926	0.955	0.96
common_support	50	donor_expression_pca	10	0.97	0.005	0.961	0.978	0.98
common_support	50	kme_multiscale@scgpt	10	0.948	0.011	0.927	0.964	0.966

2.6 Covariate sensitivity materially changes interpretation of high AUCs

The covariate audit is central to the final manuscript. In GSE174188 CD4, measured donor/sample covariates alone achieved AUC 0.846. After residualization against the frozen covariate set, scGPT mean reached AUC 0.868, donor expression PCA 0.853, and donor mean HVG 0.870. Residualized methods were not cleanly separated from the covariates-only baseline by the paired bootstrap analysis. This result does not eliminate biological disease signal; it shows that measured donor/sample variables capture a substantial component of the apparent discrimination and must be reported with the benchmark.

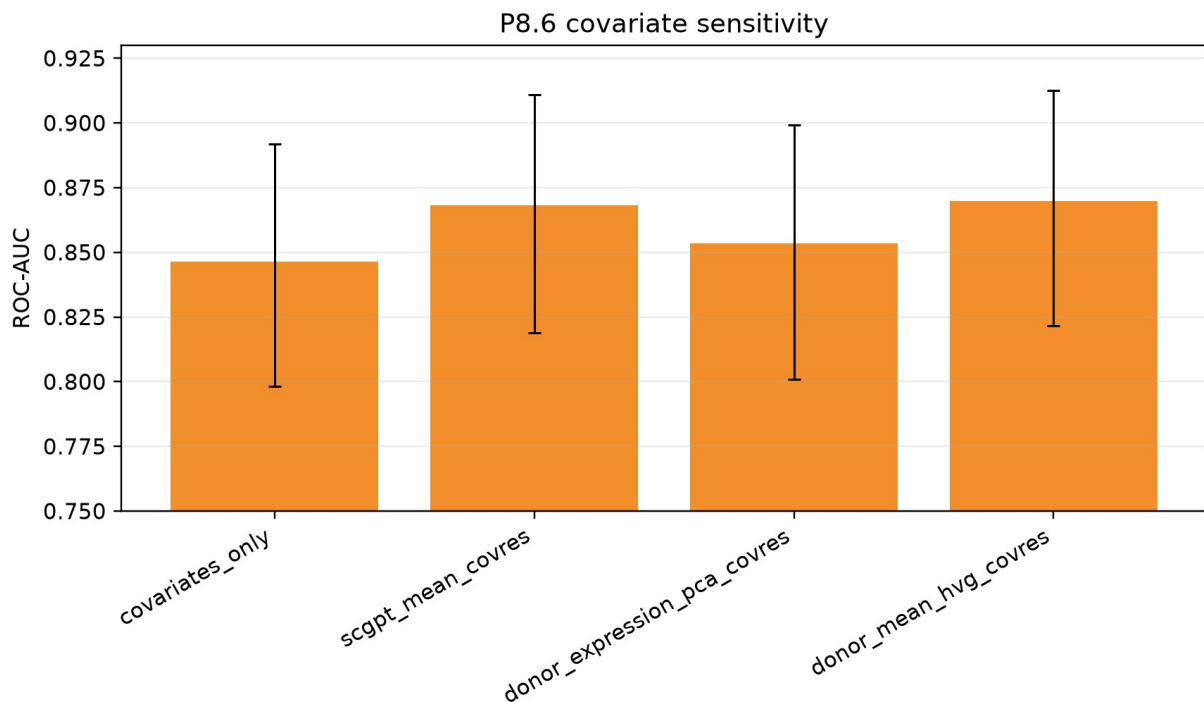


Figure 3. Covariate sensitivity.

We decomposed the covariates-only signal to make the result interpretable. Under the formal-compatible P8.6 encoding, the full covariate model reproduced the accepted result closely (AUC 0.848 versus accepted AUC 0.846). Technical depth/QC covariates alone reached AUC 0.776, processing covariates AUC 0.685, and demographic covariates AUC 0.653. The strongest single covariates were cells per donor (AUC 0.689), age from development stage (AUC 0.639), and mitochondrial percentage (AUC 0.614). A sensitivity encoding that one-hot encoded processing cohort increased covariates-only AUC to 0.930, showing that categorical handling of processing structure materially changes the audit. We therefore keep the accepted P8.6 encoding for formal claims and report one-hot processing as a sensitivity warning.

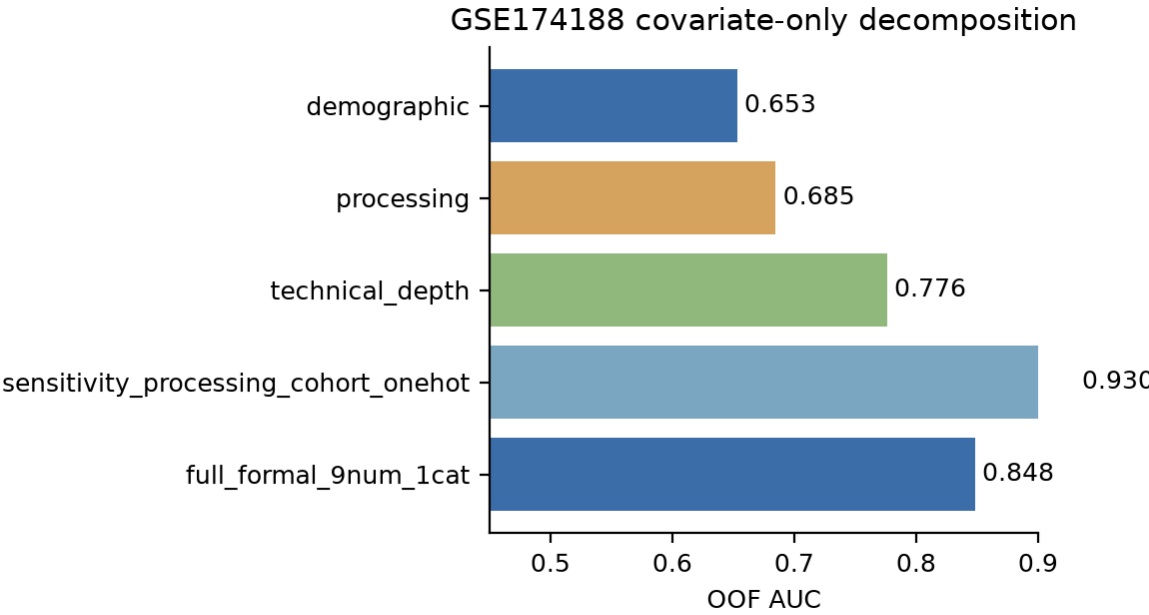


Figure 13. Covariate decomposition.

Residualized method scores were only modestly above the covariates-only baseline. The AUC increment was +0.022 for scGPT mean, +0.007 for donor expression PCA, and +0.023 for donor mean HVG; the corresponding bootstrap intervals included zero. This result motivates presenting covariate sensitivity as a primary audit endpoint rather than a supplementary afterthought.

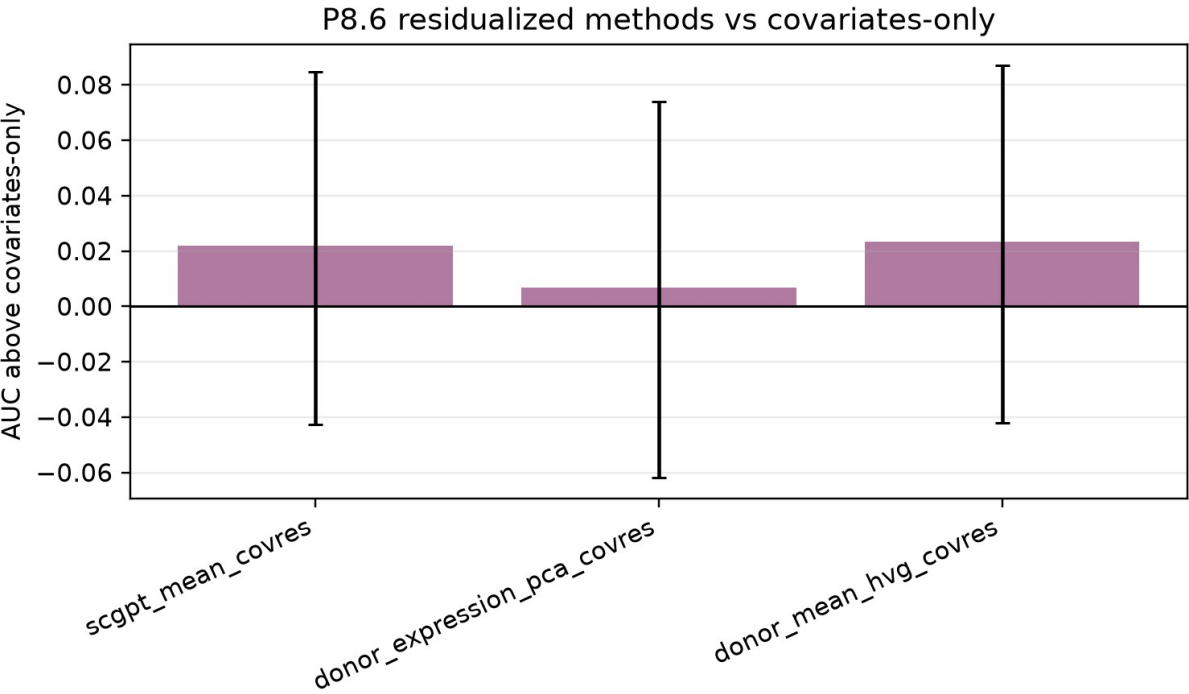


Figure 7. Residualized methods versus covariates-only.

The paired method-comparison audit further supports conservative wording near the AUC ceiling. In GSE174188 CD4, raw pseudobulk, donor expression PCA, donor mean HVG, and KME were numerically close to scGPT mean, but none of their Δ AUC intervals excluded zero after Holm correction. `focus_lite@scGPT` was significantly lower than scGPT mean in GSE174188 CD4. In GSE285773, no comparison against scGPT mean survived correction, consistent with the small 26-donor target. These results justify describing expression baselines as numerically strongest or comparable, rather than claiming broad statistical superiority across all cohorts.

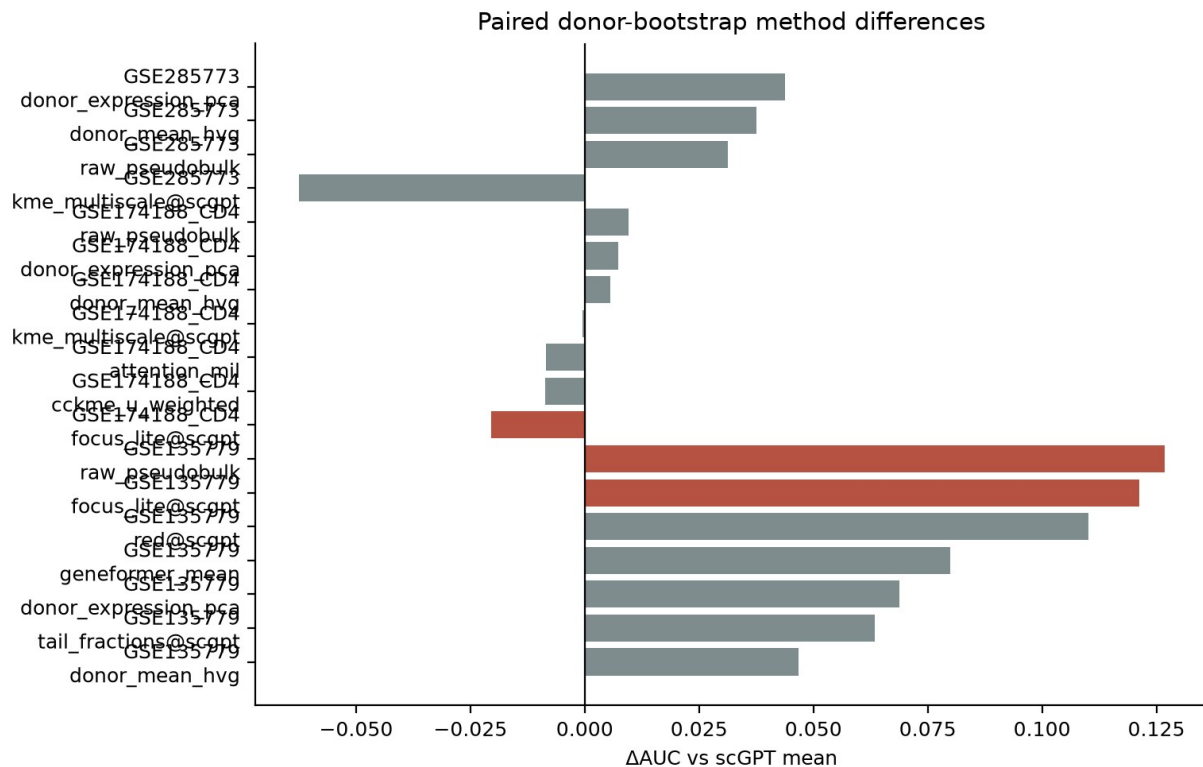


Figure 14. Paired AUC differences versus scGPT mean.

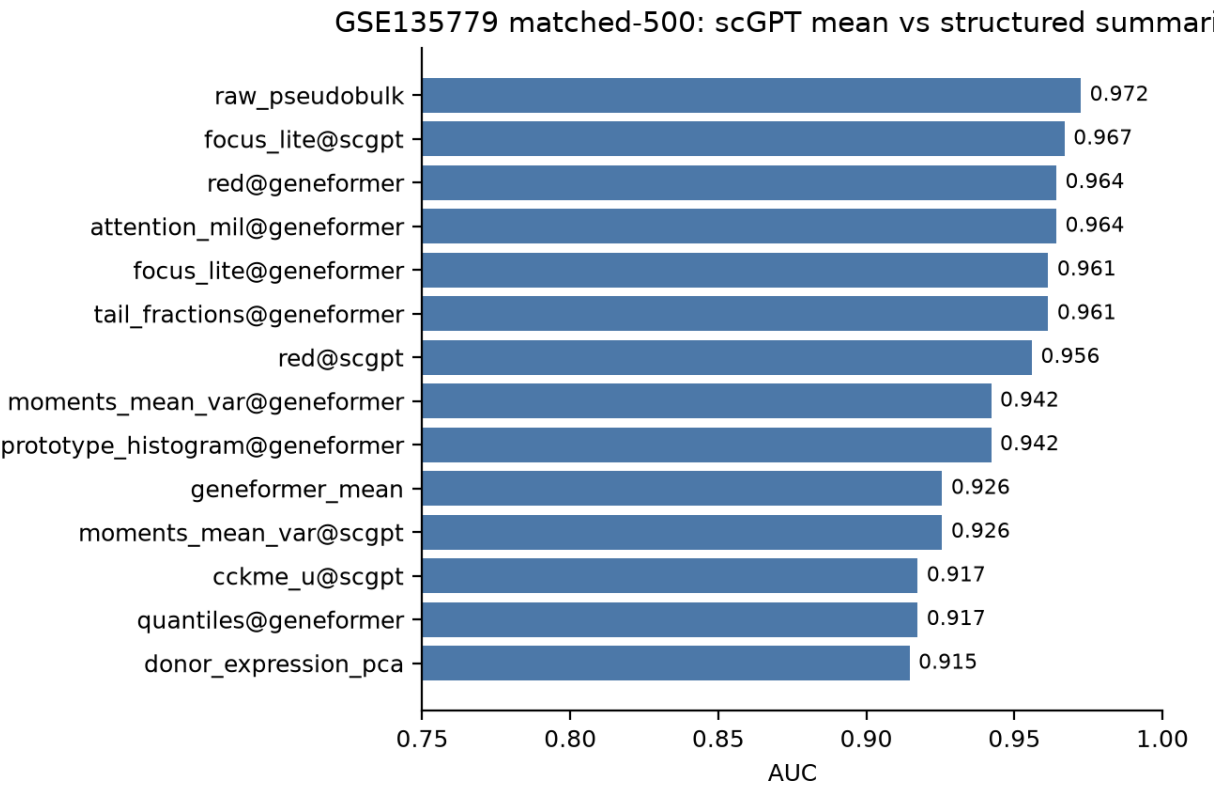


Figure 15. GSE135779 matched-500 scGPT mean context.

Table 5. Covariate sensitivity.

method_id	n_donors	n_sle	n_control	auc	pr_auc	brier	feature_dim	q025	q975
covariates_only	261	162	99	0.846	0.905	0.162	13	0.798	0.892
scgpt_mean_covres	261	162	99	0.868	0.896	0.148	13	0.819	0.911
donor_expression_pca_covres	261	162	99	0.853	0.87	0.156	13	0.801	0.899
donor_mean_hvg_covres	261	162	99	0.87	0.896	0.169	13	0.822	0.913

2.7 KME diagnostics resolved identity collapse and established a bounded interpretation

P8.7 confirmed that KME predictions are distinct from scGPT mean predictions after the V5 runner fix. Across cell-count levels, KME had zero exact prediction identity with scGPT mean, stable bandwidth values around 7.55–7.65, and effective rank approximately 2.6–2.8. The low effective rank indicates a concentrated kernel spectrum. The diagnostic result supports use of KME as a distinct distribution-level method in the benchmark and sets the interpretation boundary for mechanism or superiority claims.

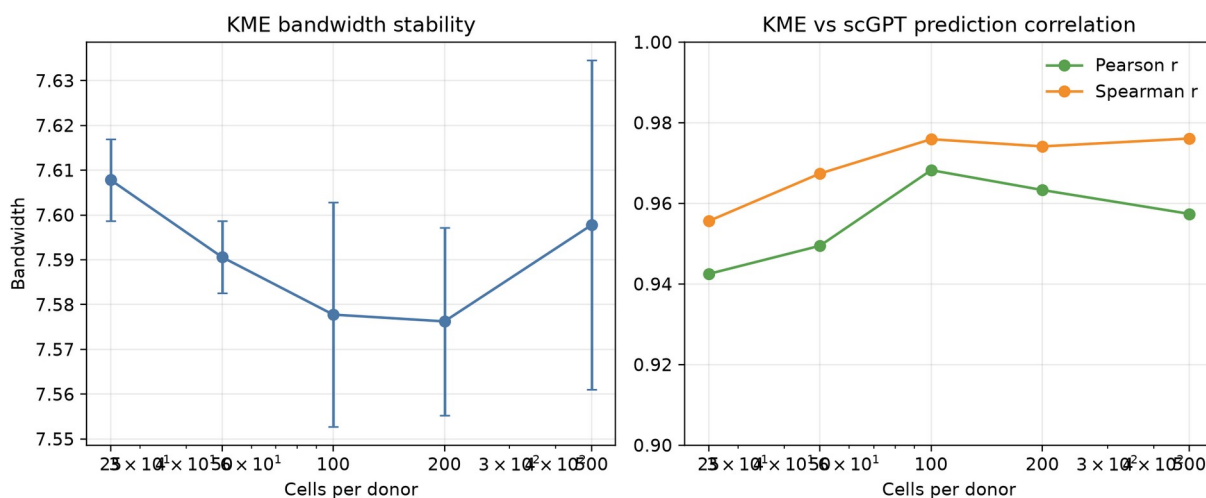


Figure 8. KME diagnostics.

Table 6. KME identity diagnostics.

level	max_abs_diff	pearson_r	spearman_r	n_exact_equal
25	0.661	0.942	0.956	0
50	0.739	0.949	0.967	0
100	0.574	0.968	0.976	0
200	0.614	0.963	0.974	0
500	0.69	0.957	0.976	0

2.8 Three-cohort method heatmap summarizes the final benchmark landscape

The three-cohort method heatmap provides a compact view of method behavior across GSE135779, GSE174188 CD4, and GSE285773. Expression methods dominate the upper range in the larger CD4 analyses, while foundation-embedding methods remain competitive and provide a reusable representation layer for distribution-level and attention-style aggregators. This pattern supports the

manuscript's main empirical message: disease signal is strong and reproducible, and the audit framework is required to determine when complex representations add value beyond expression.

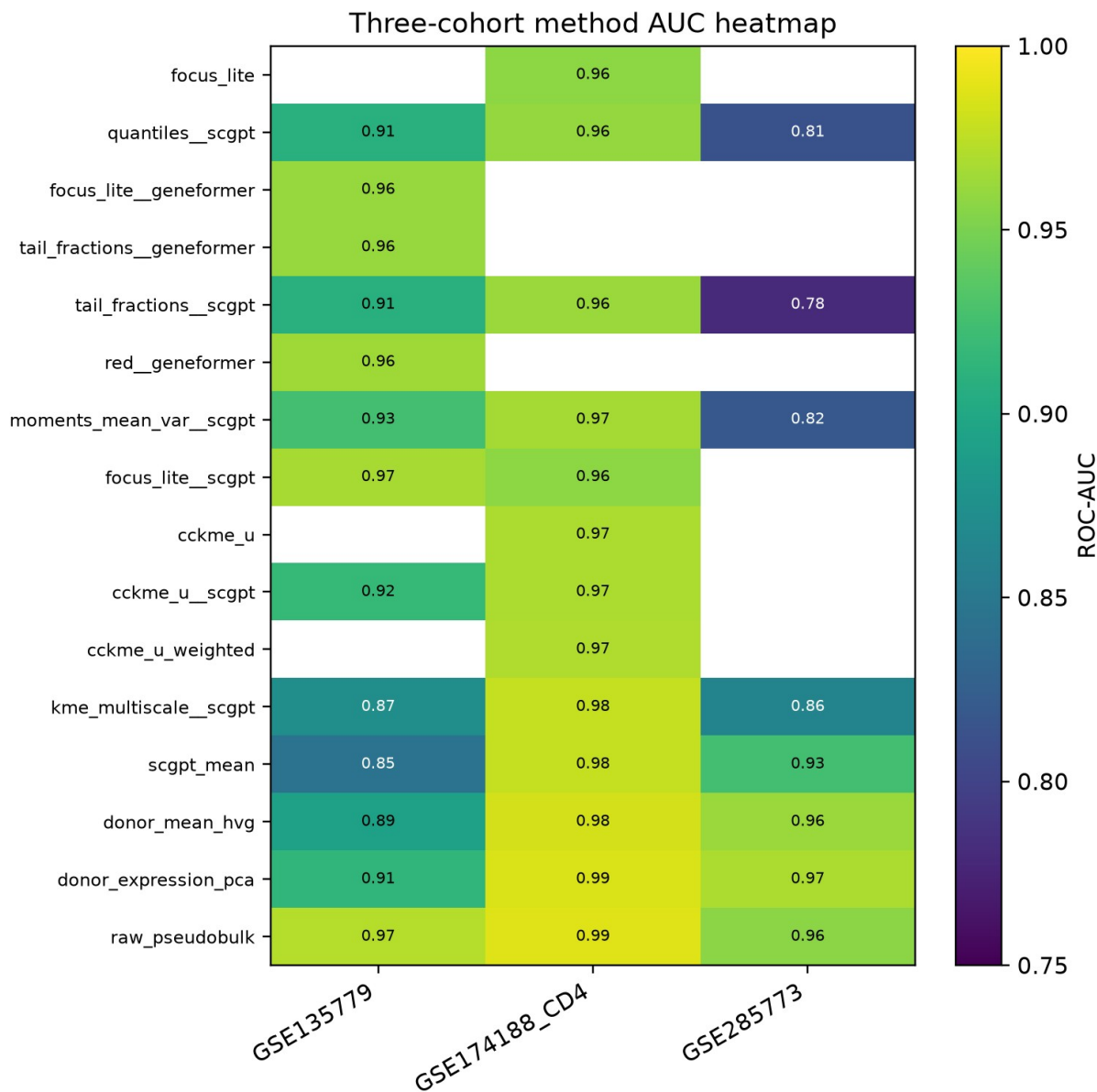


Figure 9. Three-cohort method heatmap.

2.9 Interferon and high-discrimination genes anchor the benchmark in known SLE biology

To add a direct biological anchor to the benchmark, we computed donor-level CD4 pseudobulk expression in GSE174188 after repairing gene symbols from the source h5ad file. A 20-gene interferon-stimulated gene (ISG) score was higher in SLE donors than controls and reached donor-level AUC 0.906,

Cohen $d=1.37$ and Welch $P=3.6 \times 10^{-29}$. All 20 predefined ISG genes were present in the expression matrix.

The top SLE-high donor-level genes included EPSTI1, ISG15, IFI44, IFI44L, IFI6, XAF1, IRF7 and USP18, consistent with a strong type I interferon program. Additional high-ranking genes included PSME2, LY6E, TNFRSF4 and LGALS3BP. This result supports the clinical interpretation that much of the benchmarked signal tracks a canonical lupus inflammatory axis rather than an opaque model-specific latent factor.

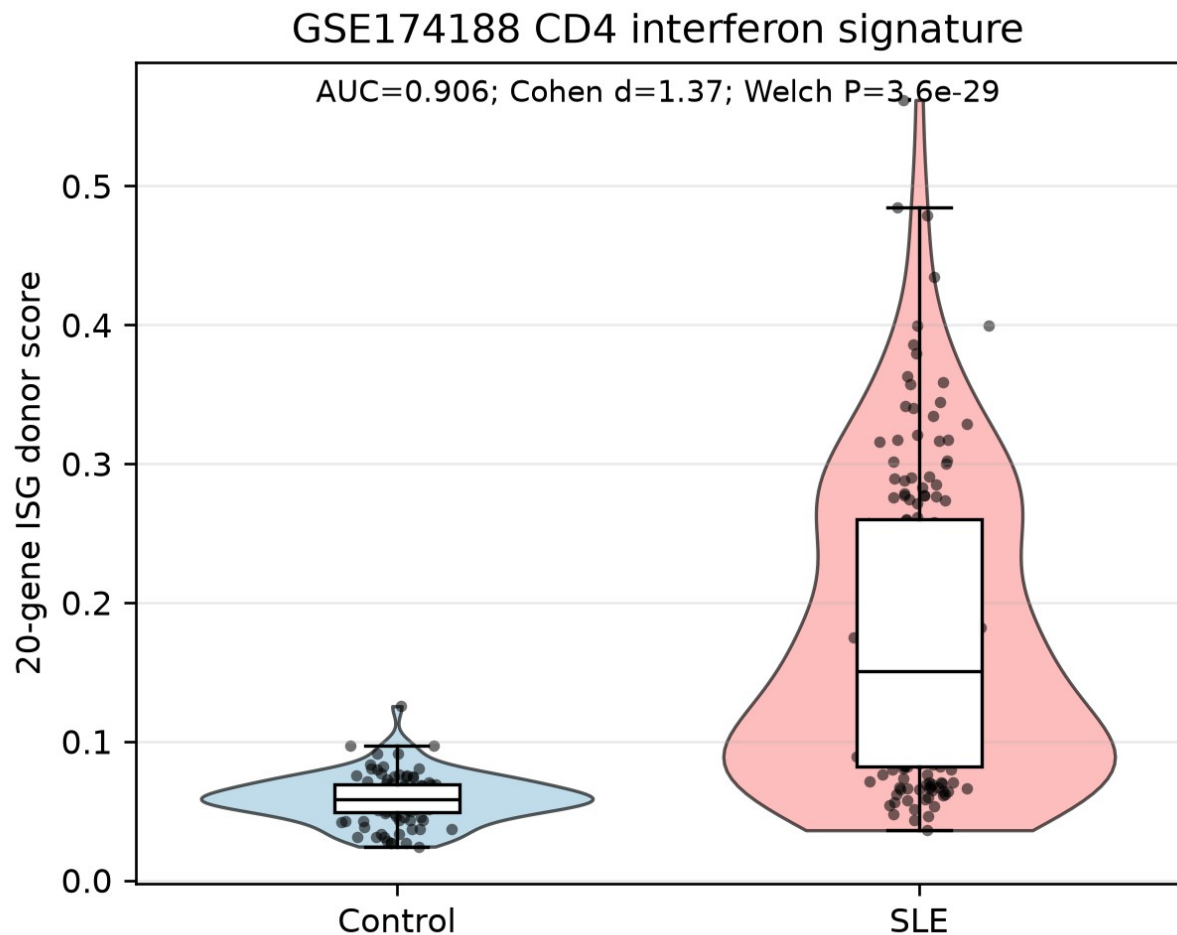


Figure 10. ISG signature validation.

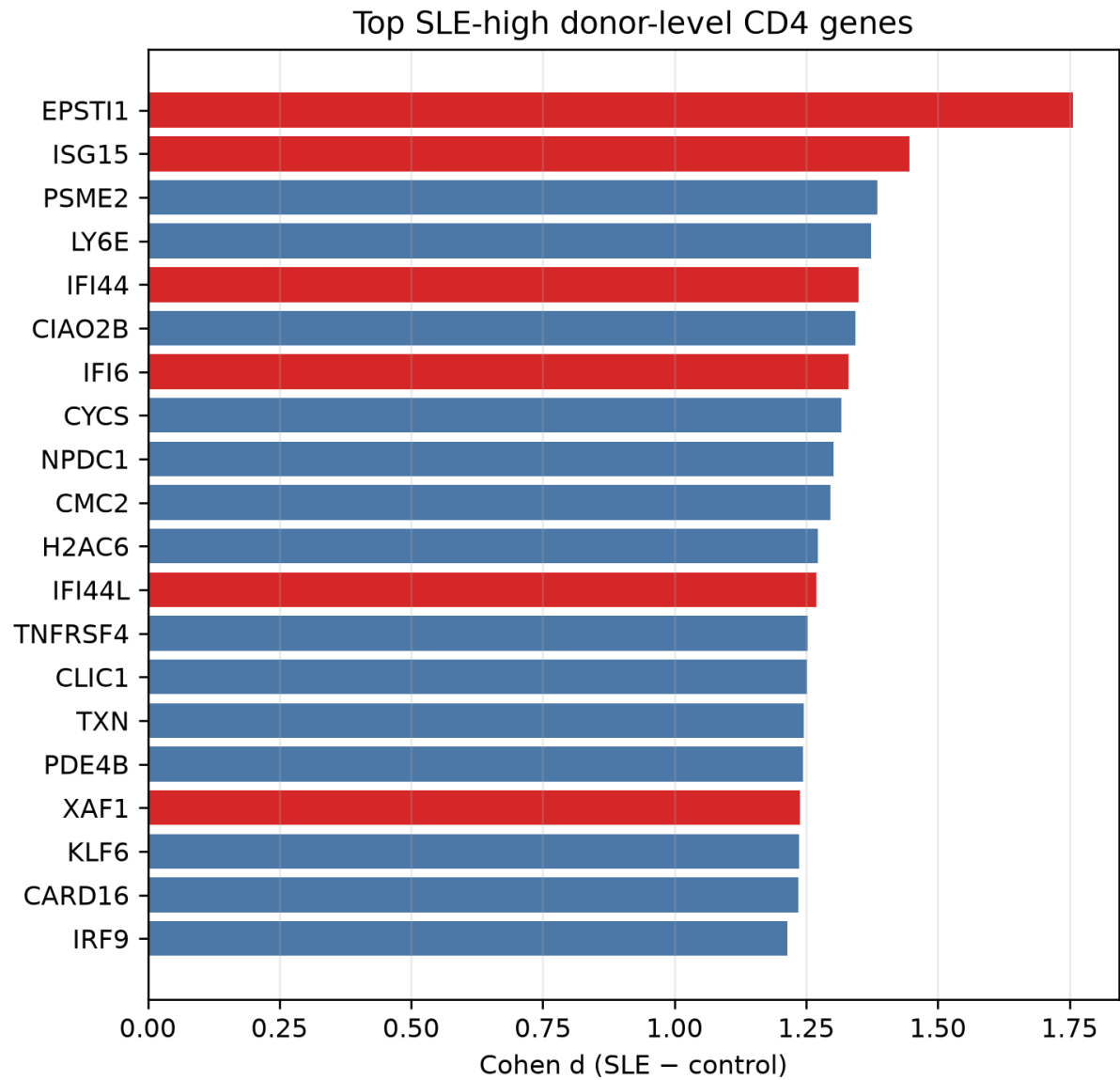


Figure 11. Top SLE-high donor-level genes.

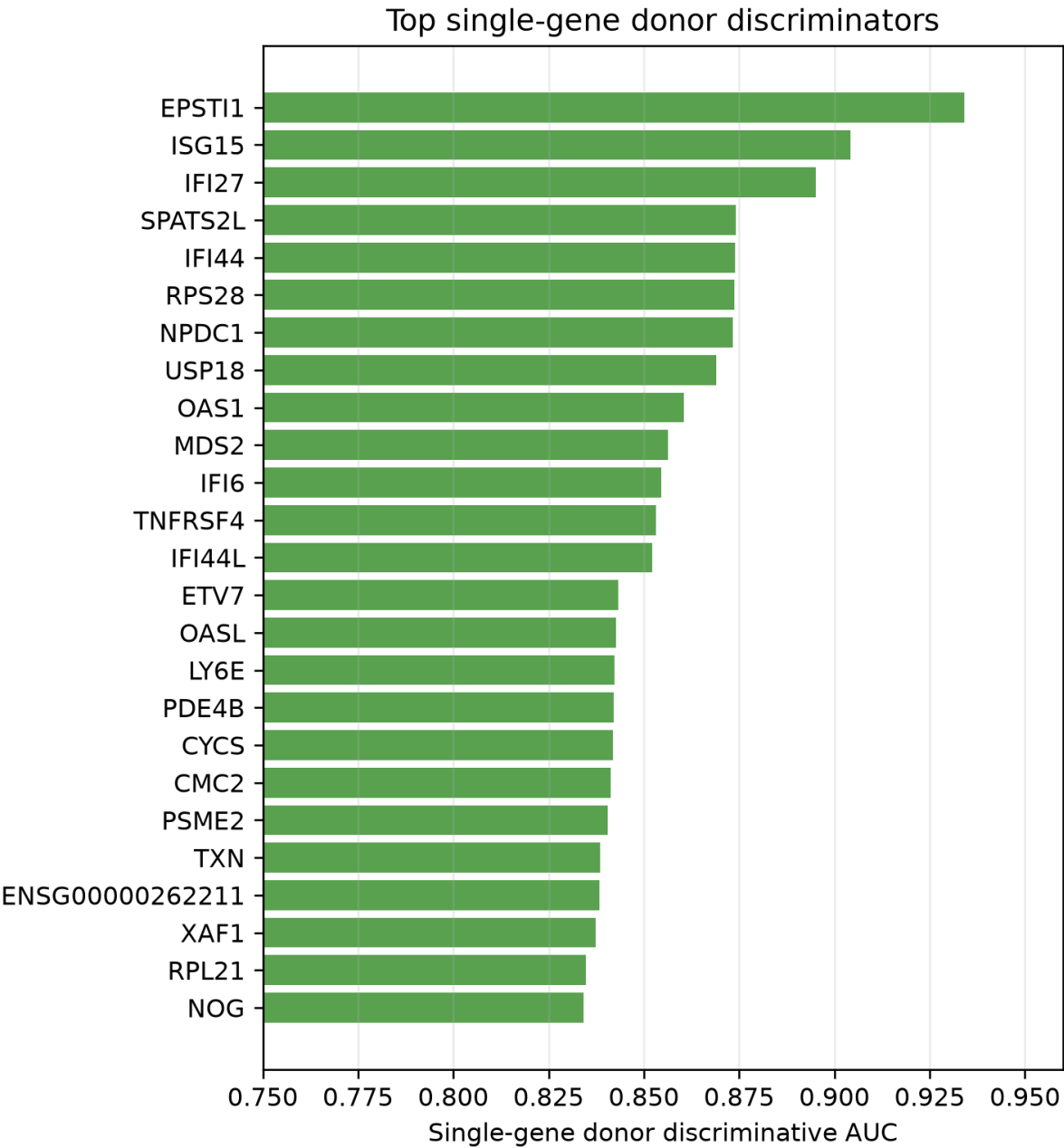


Figure 12. Top single-gene donor discriminators.

The mechanism analysis also clarifies the clinical meaning of the covariate audit. The adult GSE174188 CD4 dataset contains strong SLE/control transcriptional separation, but donor/sample covariates alone also classify well. These two findings are compatible: clinically meaningful disease biology and cohort-structured covariates can both contribute to high AUC. RheumLens therefore treats mechanism validation and covariate sensitivity as paired requirements for interpreting single-cell disease classifiers.

2.10 FOCUS and surrogate-null audits were handled by trigger logic

FOCUS results exist in P5 and P6. The earlier broad superiority phrasing was withdrawn during audit. P8.8 was therefore treated as a sensitivity-control trigger audit rather than a formal superiority test. P8.5 surrogate-null computation was skipped because KME did not show a positive incremental advantage over scGPT mean in the audited analyses. These skip decisions are recorded in the failed/skipped registry and preserve a clear boundary between completed formal evidence and exploratory method code.

3. Discussion

RheumLens establishes a practical donor-level audit standard for single-cell foundation-model disease analysis. The benchmark confirms that frozen scGPT and repaired Geneformer embeddings carry SLE-associated donor signal. It also shows that expression baselines are very strong. Raw pseudobulk, donor expression PCA, and donor mean HVG frequently match or exceed embedding-based methods in AUC.

The source-only transfer results strengthen the manuscript. They show that the donor-level signal is not restricted to within-cohort cross-validation. Transfer between pediatric and adult CD4 datasets remains informative, with expression summaries strongest in the current formal run and scGPT mean retaining clear cross-study signal.

The covariate analysis should anchor the interpretation. High AUCs in GSE174188 CD4 are partly recoverable from measured donor/sample covariates. This finding raises the technical standard for future claims. Disease-prediction performance in single-cell cohorts should be reported alongside covariate-only baselines and residualized sensitivity analyses.

The mechanism supplement gives the benchmark a biological axis. The strongest donor-level CD4 signal in GSE174188 is an interferon-centered lupus program, with EPSTI1, ISG15, IFI44, IFI44L, IFI6, XAF1, IRF7 and USP18 among the highest SLE-associated genes. This is clinically plausible because interferon activation is a recurrent SLE feature and is relevant to disease stratification and therapeutic interpretation. The pediatric-to-adult transfer results suggest that part of this transcriptional program is stable across age groups, while the covariate audit shows that clinical use would require explicit control for donor, demographic and sample-level structure.

The Geneformer repair and KME runner audit are also important. Geneformer initially appeared defective because an unknown-provenance embedding had poor disease performance. Full extraction repair restored expected performance. KME initially collapsed to mean-pooling behavior because the runner ignored registered method aggregators. The corrected V5 pipeline resolved this. These episodes

support one of the paper's main engineering claims: model-comparison papers need provenance gates and implementation audits, not only output metrics.

The main limitation is that formal claims remain retrospective and cohort-specific. GSE285773 has only 26 donors, and GSE174188 covariates strongly affect interpretation. The benchmark evaluates frozen or fixed representations rather than full task-specific fine-tuning. Clinical deployment, longitudinal prediction, treatment response, and causal biology require additional datasets and prospective designs.

4. Methods

4.1 Data sources and processed objects

The formal snapshot contains processed assets for GSE135779 matched-500, GSE174188 CD4, and GSE285773. GSE135779 matched-500 includes scGPT and Geneformer embeddings and expression matrices for 44 donors. GSE174188 CD4 includes 380,275 cells from 261 donors. GSE285773 includes 262,908 cells from 26 donors. GSE174188 feature names were repaired from the source `h5ad.var.feature_name` field before P9 expression transfer.

4.2 Donor-level evaluation

All formal classification metrics are donor-level. Cell-level representations are aggregated to donor-level features by the registered method. Cross-validation uses donor-disjoint folds. Out-of-fold predictions are pooled across held-out donors before AUC, PR-AUC, and Brier score calculation. Source-only transfer fits all preprocessing and classifiers on the source cohort and applies the fitted pipeline to the target cohort.

4.3 Method families

The benchmark includes raw pseudobulk, donor expression PCA, donor mean HVG, scGPT mean, Geneformer mean, KME/multiscale kernel mean embeddings, FOCUS-style methods, RED/tail/moment summaries, attention/MIL methods, and additional learned donor-set models. The manuscript tables emphasize methods with completed formal audit status.

4.4 Uncertainty estimates

Within-cohort sensitivity analyses use paired donor bootstrap where available. P8.2 formal permutation used 1,000 donor-label permutations for primary GSE174188 CD4 methods and empirical P-values with one-count correction. P8.3 repeated cross-validation used 30 stratified donor-level repeats to quantify fold-resampling stability. P9 source-only transfer reports AUC intervals using a Hanley-McNeil large-sample approximation and PR-AUC intervals using stratified donor bootstrap. Bootstrap distributions,

permutation distributions, repeated-CV summaries and out-of-fold predictions are archived with SHA256 manifests.

4.5 Covariate sensitivity

P8.6 used a frozen covariate set derived from authoritative donor metadata. The final V3.1 analysis includes covariates-only prediction and residualized versions of scGPT mean, donor expression PCA, and donor mean HVG. The sensitivity analysis is interpreted as an audit of robustness to measured covariates.

4.6 Mechanism-oriented gene analysis

For GSE174188 CD4, log-normalized cell-level expression was aggregated to donor-level pseudobulk means after gene-symbol repair from the source `h5ad var.feature_name` field. Donor-level SLE-control gene differences were summarized by mean difference, Welch t statistic, Benjamini-Hochberg FDR, Cohen d and single-gene donor AUC. Duplicate gene symbols were collapsed by retaining the most discriminative feature. A predefined 20-gene ISG score was computed as the donor-level mean of IFI44L, IFI44, IFI27, ISG15, MX1, MX2, OAS1, OAS2, OAS3, OASL, IFIT1, IFIT2, IFIT3, RSAD2, SIGLEC1, STAT1, IRF7, USP18, XAF1 and EPSTI1.

4.7 Reproducibility

All formal outputs are tracked by SHA256 manifests in the open repository and the accompanying archival data package. The manuscript asset directory contains plot-ready tables, vector/raster figures, and audit reports. GSE174188 feature-name repair is documented as a derived metadata asset generated from the source `h5ad var.feature_name` field.

5. Data and code availability

Source datasets are publicly available through GEO accessions GSE135779, GSE174188 and GSE285773 or their associated source repositories. The RheumLens code, manuscript assets, audit scripts, fold definitions, plot-ready tables and SHA256 manifests are maintained at <https://github.com/LightChainr/rheumlens>; a DOI-tagged archival snapshot will be minted before submission. Processed donor-level outputs, out-of-fold predictions, bootstrap and permutation distributions, repeated-CV summaries, covariate sensitivity outputs, source-only transfer predictions and figure-ready tables will be deposited in an archival data release with the same SHA256 manifest. Large cell-level matrices and foundation-model embeddings will be distributed as separate archival objects or regeneration recipes, depending on repository size constraints and source-data redistribution terms.

6. Current manuscript figure legends

Figure 1. Top benchmark AUCs across matched-500 and GSE174188 CD4 analyses. Horizontal bars show leading method ROC-AUC values. The figure highlights high donor-level disease discrimination across expression and foundation-embedding methods.

Figure 2. Source-only CD4 cross-study transfer. Models were trained on one cohort and evaluated on the other without target-label fitting. Error bars show AUC intervals.

Figure 3. Covariate sensitivity in GSE174188 CD4. Covariates-only and residualized method AUCs are shown with bootstrap intervals. Measured covariates account for a substantial component of donor-level discrimination.

Figure 4. Repeated donor-level cross-validation stability. Bars show mean AUC across 30 repeated donor-level cross-validation runs; error bars show the standard deviation. Black points show the corresponding fixed-fold P8.2 reference values.

Figure 5. Formal donor-label permutation null distributions. Histograms show 1,000 donor-label permutation AUCs for donor expression PCA and kme_multiscale@scGPT. Red lines show observed AUCs. Dashed black lines show the 97.5th percentile of each null distribution.

Figure 6. Cell-count sensitivity. Common-support donor sampling was evaluated at 25, 50, 100, 200 and 500 cells per donor. Points show mean AUC across 10 repeats; error bars show 2.5th–97.5th percentile intervals.

Figure 7. Residualized methods versus covariates-only. Bars show AUC increment over the covariates-only baseline for residualized scGPT mean, donor expression PCA and donor mean HVG. Error bars show bootstrap intervals.

Figure 8. KME diagnostics. Left: bandwidth stability across cell-count levels. Right: Pearson and Spearman correlations between KME and scGPT mean donor predictions after identity-collapse repair.

Figure 9. Three-cohort method heatmap. Heatmap summarizing method AUCs across completed cohorts and modalities in the final P6/P7 integrated benchmark table.

Figure 10. ISG signature validation. Donor-level 20-gene interferon-stimulated gene score in GSE174188 CD4, stratified by control and SLE labels. The score achieved AUC 0.906, Cohen $d=1.37$ and Welch $P=3.6 \times 10^{-29}$.

Figure 11. Top SLE-high donor-level CD4 genes. Bars show Cohen d for the top SLE-high genes from GSE174188 CD4 donor-level pseudobulk expression. Interferon-associated genes are highlighted.

Figure 12. Top single-gene donor discriminators. Bars show single-gene donor-level discriminative AUC for the top genes in GSE174188 CD4.

Figure 13. Covariate decomposition. Bars show GSE174188 CD4 covariates-only AUC under the formal-compatible encoding, grouped covariate models, and the one-hot processing-cohort sensitivity encoding.

Figure 14. Paired AUC differences versus scGPT mean. Bars show donor-bootstrap Δ AUC for selected methods compared with scGPT mean within each cohort. Red bars indicate Holm-adjusted $P < 0.05$.

Figure 15. GSE135779 matched-500 scGPT mean context. Bars show the top GSE135779 matched-500 method AUCs, highlighting that several structured or distributional summaries exceeded scGPT mean in this cohort.

Figure 16. P9 structured source-only transfer extension. Bars show source-only transfer AUC for non-neural structured methods in both CD4 transfer directions. Neural MIL transfer was not included in this extension because it requires a separate pre-registered source-target training protocol.

7. Supplementary material plan

- Supplementary Table S1: full method summary for P5 matched-500.
- Supplementary Table S2: full method summary for P6 GSE174188 CD4.
- Supplementary Table S3: integrated P6 all-cohort method summary.
- Supplementary Table S4: P8.3 repeated-CV summary.
- Supplementary Table S5: P8.4 cell-count sensitivity full table.
- Supplementary Table S6: P8.6 covariate summary.
- Supplementary Table S7: P8.6 residualized-method increment over covariates-only.
- Supplementary Table S8: P8.7 KME diagnostics.
- Supplementary Table S9: P9 source-only transfer summary.
- Supplementary Table S10: P9 source-only transfer donor-level predictions.
- Supplementary Table S11: P8.2 formal permutation summary.
- Supplementary Table S12: GSE174188 CD4 donor-level gene discrimination table.
- Supplementary Table S13: top SLE-high and control-high donor-level genes.
- Supplementary Table S14: donor-level ISG scores and ISG summary.
- Supplementary Table S15: GSE174188 covariate decomposition and encoding sensitivity.
- Supplementary Table S16: paired donor-bootstrap Δ AUC with Holm correction.
- Supplementary Table S17: GSE135779 matched-500 scGPT mean context table.
- Supplementary Table S18: P9 structured source-only transfer extension.

- Supplementary Methods: computational environments, data provenance, donor-disjoint folds, Geneformer provenance repair, method registry, P8 audit gates, superseded analyses, KME V5 runner audit, FOCUS trigger audit, source-only transfer, and claim boundary.

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